45-Day DSA Mastery Roadmap

Comprehensive LeetCode Problem Plan • Curated from Blind 75 & NeetCode 150

45 Days | 150+ Problems | Structured Preparation

Target Goal: Complete Data Structures & Algorithms Mastery

Core Topics: Arrays, Graphs, DP, Trees & Search

Phase 1: Arrays, Two Pointers & Sliding Window

Day 1: Arrays & Hashing — FUNDAMENTALS

☐	Two Sum	EASY
☐	Contains Duplicate	EASY
☐	Best Time to Buy and Sell Stock	EASY

Day 2: Arrays & Hashing (Advanced) — HASHING & FREQUENCY

☐	Valid Anagram	EASY
☐	Group Anagrams	MEDIUM
☐	Top K Frequent Elements	MEDIUM
☐	Product of Array Except Self	MEDIUM

Day 3: Sliding Window — SUBSTRINGS

☐	Longest Substring Without Repeating Characters	MEDIUM
☐	Longest Repeating Character Replacement	MEDIUM

Day 4: Two Pointers — LINEAR SCANNING

☐	Valid Palindrome	EASY
☐	3Sum	MEDIUM
☐	Container With Most Water	MEDIUM

Day 5: Two Pointers (Hard) — ADVANCED POINTERS

☐	Trapping Rain Water	HARD

Day 6: Revision & Review

Review Days 1–5 problems. Re-solve missed problems and analyze time/space complexity.

Phase 2: Stacks, Binary Search & Linked Lists

Day 7: Stack Basics — DATA STRUCTURES

☐	Valid Parentheses	EASY
☐	Min Stack	MEDIUM

Day 8: Sliding Window (Hard) — ADVANCED WINDOW

- [ ] Minimum Window Substring — HARD
- [ ] Permutation in String — MEDIUM

Day 9: Sliding Window Maximum — MONOTONIC QUEUE

- [ ] Sliding Window Maximum — HARD

Day 10: Binary Search Intro — SEARCHING

- [ ] Binary Search — EASY
- [ ] Search a 2D Matrix — MEDIUM
- [ ] Koko Eating Bananas — MEDIUM

Day 11 & 12: Advanced Stack — MONOTONIC STACK

- [ ] Evaluate Reverse Polish Notation — MEDIUM
- [ ] Daily Temperatures — MEDIUM
- [ ] Car Fleet — MEDIUM
- [ ] Largest Rectangle in Histogram — HARD

Day 13 – 15: Binary Search Mastery — ROTATED & SORTED

- [ ] Find Minimum in Rotated Sorted Array — MEDIUM
- [ ] Search in Rotated Sorted Array — MEDIUM
- [ ] Time Based Key-Value Store — MEDIUM
- [ ] Median of Two Sorted Arrays — HARD

Day 16 – 18: Linked List — POINTERS & MANIPULATION

- [ ] Reverse Linked List — EASY
- [ ] Merge Two Sorted Lists — EASY
- [ ] Reorder List — MEDIUM
- [ ] Remove Nth Node From End of List — MEDIUM
- [ ] Copy List with Random Pointer — MEDIUM
- [ ] Linked List Cycle — EASY
- [ ] Find the Duplicate Number — MEDIUM

Day 19 & 20: Mid-Term Revision

Re-visit Linked Lists, Monotonic Stacks, and Rotated Binary Search problems.

Phase 3: Strings, Bit Manipulation & Trees

Day 21: Strings — PALINDROMES & ENCODING

☐	Longest Palindromic Substring	MEDIUM
☐	Palindromic Substrings	MEDIUM
☐	Encode and Decode Strings	MEDIUM

Day 22: Bit Manipulation — BITWISE OPERATORS

☐	Single Number	EASY
☐	Number of 1 Bits	EASY
☐	Counting Bits	EASY
☐	Reverse Bits	EASY
☐	Missing Number	EASY

Day 23 – 25: Binary Trees (DFS & BFS) — TREE TRAVERSALS

☐	Invert Binary Tree	EASY
☐	Maximum Depth of Binary Tree	EASY
☐	Same Tree	EASY
☐	Subtree of Another Tree	EASY
☐	Lowest Common Ancestor of a BST	MEDIUM
☐	Binary Tree Level Order Traversal	MEDIUM

Day 26 & 27: Binary Search Trees & Advanced Trees — BST PROPERTIES

☐	Validate Binary Search Tree	MEDIUM
☐	Kth Smallest Element in a BST	MEDIUM
☐	Construct Tree from Preorder & Inorder Traversal	MEDIUM
☐	Binary Tree Maximum Path Sum	HARD

Day 28: Tries (Prefix Tree) — STRING TREES

☐	Implement Trie (Prefix Tree)	MEDIUM
☐	Design Add and Search Words Data Structure	MEDIUM
☐	Word Search II	HARD

Day 29: Tree Revision

Review DFS/BFS tree recursions, tree construction algorithms, and Trie traversals.

Phase 4: Heaps & Graphs

Day 30: Priority Queue / Heaps — MIN/MAX HEAPS

☐	Kth Largest Element in a Stream	EASY
☐	Last Stone Weight	EASY
☐	K Closest Points to Origin	MEDIUM
☐	Task Scheduler	MEDIUM
☐	Find Median from Data Stream	HARD

Day 31 – 33: Graphs (BFS & DFS) — GRID & CONNECTIVITY

☐	Number of Islands	MEDIUM
☐	Max Area of Island	MEDIUM
☐	Pacific Atlantic Water Flow	MEDIUM
☐	Course Schedule & Course Schedule II	MEDIUM
☐	Rotting Oranges	MEDIUM

Day 34: Union Find & Shortest Path — ADVANCED GRAPHS

☐	Redundant Connection (Union Find)	MEDIUM
☐	Number of Connected Components	MEDIUM
☐	Network Delay Time (Dijkstra's)	MEDIUM
☐	Cheapest Flights Within K Stops	MEDIUM

Day 35: Graph Revision

Review Topological Sort (Kahn's Algorithm), Union Find, and Shortest Path algorithms.

Phase 5: Backtracking & Dynamic Programming

Day 36 – 38: Backtracking — COMBINATORICS

☐	Subsets & Subsets II	MEDIUM
☐	Combination Sum & Combination Sum II	MEDIUM
☐	Permutations	MEDIUM
☐	Word Search	MEDIUM
☐	Letter Combinations of a Phone Number	MEDIUM
☐	N-Queens	HARD

Day 39 – 41: 1D Dynamic Programming

SUBPROBLEMS & MEMOIZATION

☐	Climbing Stairs	EASY
☐	Min Cost Climbing Stairs	EASY
☐	House Robber & House Robber II	MEDIUM
☐	Coin Change	MEDIUM
☐	Maximum Subarray	MEDIUM
☐	Word Break	MEDIUM
☐	Longest Increasing Subsequence	MEDIUM

Day 42: DP 1D Revision

Consolidate state transitions and base cases for 1D Dynamic Programming patterns.

Day 43 & 44: 2D Dynamic Programming

GRID & SEQUENCE DP

☐	Unique Paths	MEDIUM
☐	Longest Common Subsequence	MEDIUM
☐	Best Time to Buy & Sell Stock with Cooldown	MEDIUM
☐	Target Sum	MEDIUM
☐	Edit Distance	HARD
☐	Burst Balloons	HARD
☐	Regular Expression Matching	HARD

Day 45: Final Mock Test & Mixed Review

INTERVIEW PREP

☐	Random Mixed Problem Set (2 Mediums, 1 Hard)	MOCK